

Shafizur Rahman Seeam

Ph.D. Student in Computing and Information Sciences
Department of Cybersecurity, Rochester Institute of Technology

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Research Interests

Privacy-Enhancing Technologies • Differential Privacy • AI Agent Privacy

Education

Rochester Institute of Technology, Rochester, NY
Ph.D. in Computing and Information Sciences

Jan. 2023 – Present
Expected May 2027

- Advisor: Dr. Yidan Hu
- Dissertation: *Beyond Isolated Decisions: Advancing Dependence-Aware User-Side Privacy Mechanisms*

University of Dhaka, Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering

Jan. 2015 – Dec. 2019

Publications

Under Review

- [1] **Shafizur Rahman Seeam**, Zhengxiong Li, Zhiyuan Yu, Yimin Chen, and Yidan Hu. **PrivScope: Task-scoped Disclosure Control for Information-seeking Hybrid Agentic Systems**. *Manuscript under review*, 2026.

Peer-Reviewed Publications

- [2] **Shafizur Rahman Seeam**, Ye Zheng, Zhengxiong Li, and Yidan Hu. **PrivAR: Client-Side Privacy Framework for Real-Time Location-Based Augmented Reality**. *46th IEEE International Conference on Distributed Computing Systems (ICDCS)*, Seoul, South Korea, 2026.
- [3] **Shafizur Rahman Seeam**, Ye Zheng, and Yidan Hu. **Frequency Estimation of Correlated Multi-attribute Data under Local Differential Privacy**. *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2026.
- [4] Ye Zheng, **Shafizur Rahman Seeam**, Yidan Hu, Rui Zhang, and Yanchao Zhang. **Locally Differentially Private Frequency Estimation via Joint Randomized Response**. *Proceedings on Privacy Enhancing Technologies (PoPETs)*, 2025.
- [5] **Shafizur Rahman Seeam**, Zhengxiong Li, and Yidan Hu. **A Black-Box Approach for Quantifying Leakage of Trace-Based Correlated Data**. *7th International Workshop on Physics Embedded AI Solutions in Mobile Computing (PICASSO)*, *ACM MobiCom*, 2024.
- [6] Swapnil Sayan Saha, **Shafizur Rahman Seeam**, Mosabber Uddin Ahmed, and Subrata Kumar Aditya. **Ensuring Cybersecure Telemetry and Telecommand in Small Satellites: Recent Trends and Empirical Propositions**. *IEEE Aerospace and Electronic Systems Magazine*, vol. 34, no. 8, pp. 34–49, 2019.
- [7] Swapnil Sayan Saha, **Shafizur Rahman Seeam**, Zarin Rezwana Ridita Haque, Tahera Hossain, Sozo Inoue, and Md Atiqur Rahman Ahad. **Position Independent Activity Recognition Using Shallow Neural Architecture and Empirical Modeling**. *ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC)*, 2019.
- [8] Swapnil Sayan Saha, **Shafizur Rahman Seeam**, Miftahul Jannat Rasna, Tahera Hossain, Sozo Inoue, and Md Atiqur Rahman Ahad. **Supervised and Neural Classifiers for Locomotion Analysis**. *ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp)*, pp. 1563–1570, 2018.
- [9] Swapnil Sayan Saha, **Shafizur Rahman Seeam**, Miftahul Jannat Rasna, Tarek Bin Zahid, A.K.M. Mahfuzul Islam, and Md Atiqur Rahman Ahad. **Feature Extraction, Performance Analysis, and System Design Using the DU Mobility Dataset**. *IEEE Access*, vol. 6, pp. 44776–44786, 2018.
- [10] Swapnil Sayan Saha, **Shafizur Rahman Seeam**, Miftahul Jannat Rasna, A.K.M. Mahfuzul Islam, and Md Atiqur Rahman Ahad. **DU-MD: An Open-Source Human Action Dataset for Ubiquitous Wearable Sensors**. *7th International Conference on Informatics, Electronics & Vision (ICIEV)*, Kitakyushu, Japan, pp. 567–572, 2018. **Best Paper Award Candidate**.

Research Experience

Graduate Research Assistant — Rochester Institute of Technology

Jan. 2023 – Present

Advisor: Dr. Yidan Hu

- **Privacy-preserving agentic systems:** Developed PRIVSCOPE, a user-side disclosure-control mechanism for hybrid local–cloud agents that determines what information a delegated task requires and releases it at the least-specific sufficient level.
- **Differential privacy:** Developed Corr-RR, a two-phase mechanism that exploits inter-attribute dependence to improve multi-attribute frequency estimation while preserving record-level local differential privacy.
- **Location privacy:** Developed PRIVAR, a fully client-side framework for continuous, latency-sensitive location reporting that reduces per-location distortion and limits trajectory exposure without requiring server-side changes.

Teaching Experience

Instructor (Upcoming) — Authentication and Security Models (CSEC 472)

Spring 2027

Rochester Institute of Technology, Rochester, NY

- Appointed to serve as the primary instructor for approximately 35 students, with responsibility for course design, instruction, assessment, and student mentoring.

Graduate Teaching Assistant

Aug. 2024 – Present

Rochester Institute of Technology, Rochester, NY

- *Advanced Data Privacy (CSEC 721)* — Spring 2026; contributed to course and project design, graded student work, and mentored students throughout their projects.
- *Authentication and Security Models (CSEC 472)* — Fall 2024, Spring 2025, Fall 2025, and Fall 2026; contributed to course design, created and graded laboratory assignments, and evaluated course projects and other submissions.
- *Graduate Seminar in Computing Security (CSEC 759)* — Spring 2025.

Work Experience

Co-Founder & Director — Orboroi AB, Stockholm, Sweden

Jul. 2021 – Present

- Lead Bangladesh operations, including team coordination, client engagement, and project delivery.
- Oversee financial operations, including cash flow, payments, and expense control.
- Deploy and maintain the company website, database, CVAT platform, and Linux servers.

Awards & Honors

- **PETS 2026 Student Travel Stipend Award** (\$1,000), 2026.
- **Best Paper Award Candidate**, ICIEV, Kitakyushu, Japan, 2018.
- **3rd Place**, Digital Khichuri Challenge, UNDP & ICT Division, Bangladesh, 2017.
- **1st and 2nd Place**, Bangladesh Electronics Olympiad Project Competition, 2015 and 2017.
- **Champion, Project Showcasing (Junior)**, EEE Day, BUET, 2015 and 2016.
- **Runner-Up, Project Showcasing**, Mecceleration, IUT, 2015.

Academic Service

- Judge, GENIUS Olympiad international high-school project competition, 2023 and 2024.

Technical Skills

Programming: Python, Kotlin, SQL, MATLAB

Frameworks & Platforms: TensorFlow, LangChain, Ollama, Android, Node.js, MongoDB, CVAT

Systems & Tools: Linux, Git, L^AT_EX